ICSE CLASS 10 BIOLOGY — COMPLETE REVISION NOTES

CHAPTER 1: CELL DIVISION AND CELL CYCLE

YouTube Lecture: https://www.youtube.com/live/dLZxcbB9crs?si=Ot3H6KW22V9zLVA7

1. The Cell Cycle
The cell cycle is the sequence of events from one cell division to the next. It consists of two main phases.

A. Interphase (Preparatory Phase)
The cell grows in size and prepares for division. It is divided into three stages.
G_1 Phase: The cell grows. Proteins and RNA are synthesised.
S Phase: DNA replication occurs. Each chromosome duplicates to form two sister chromatids held together at the centromere.
G_2 Phase: Further growth happens. Proteins required for cell division are synthesised.

B. M Phase (Mitotic Phase)
This is the phase of actual cell division. It includes mitosis (nuclear division) and cytokinesis (cytoplasmic division).

2. Mitosis (Equational Division)
Mitosis occurs in somatic (body) cells for growth, repair and replacement. The chromosome number remains the same (2n to 2n). It produces two genetically identical daughter cells.

Stages of Mitosis
Prophase: Chromosomes condense and become visible as two chromatids joined at the centromere. The nuclear membrane and nucleolus disappear. Spindle fibres begin to form.
Metaphase: Chromosomes align at the equator of the cell. Spindle fibres attach to the centromeres.
Anaphase: The centromere of each chromosome splits. The sister chromatids separate and move towards opposite poles of the cell.
Telophase: Chromosomes reach the poles and decondense. The nuclear membrane and nucleolus reappear. Spindle fibres disappear. This is followed by cytokinesis, the division of cytoplasm.

3. Meiosis (Reduction Division) — Basic Understanding
Meiosis takes place in germ cells to form gametes (sperm and ovum). The chromosome number is halved (2n to n). It involves two successive divisions: Meiosis I and Meiosis II.

Key Concepts in Meiosis
Homologous Chromosomes: Pairs of similar chromosomes, one from each parent, that come together during meiosis.
Crossing Over: Homologous chromosomes exchange segments of genetic material during Meiosis I. This creates new gene combinations.
Significance of Meiosis:
It maintains the chromosome number constant across generations.

Crossing over during meiosis leads to genetic variations, which are important for evolution.

4. Differences Between Mitosis and Meiosis
Mitosis:
Occurs in somatic cells.
Chromosome number remains the same (2n to 2n).
Produces two daughter cells.
No crossing over.
Meiosis:
Occurs in germ cells.
Chromosome number is halved (2n to n).
Produces four daughter cells.
Crossing over takes place.

5. Structure of Chromosome
A chromosome is a thread like structure present in the nucleus, made of DNA and proteins.
Chromatin: The uncoiled, thread like form of DNA present during interphase.
Chromatid: Each of the two identical strands of a duplicated chromosome.
Centromere: The constricted region that holds the two sister chromatids together.
Gene: A specific segment of DNA that codes for a particular protein or characteristic.
DNA Structure: DNA is a double helix molecule (like a twisted ladder) composed of units called nucleotides.

IMPORTANT QUESTIONS: CELL DIVISION AND CELL CYCLE

Q1. Name the phases of the cell cycle and describe the events occurring in each phase of interphase.

Q2. Draw a labelled diagram of any one stage of mitosis and describe the key events occurring in that stage.

Q3. State three major differences between mitotic and meiotic division.

Q4. Explain the significance of meiosis in living organisms.

Q5. Define the terms: Chromatin, Chromatid, Centromere, Gene.

Q6. What is crossing over? During which stage of meiosis does it occur and what is its significance?

Q7. Why is mitosis called equational division?

Q8. Name the stage of mitosis where chromosomes align at the equator.

Q9. Give one difference between interphase and M phase.

Q10. What is cytokinesis?

CHAPTER 2: GENETICS

YouTube Lecture: https://www.youtube.com/live/BcFE9Fg7r1Y?si=9M02MmFdtEmpWD5W

1. Mendel's Laws of Inheritance
Gregor Mendel conducted experiments on garden pea plants and proposed three fundamental laws of inheritance.

A. Law of Dominance
When two contrasting alleles are present together (heterozygous condition), the allele that expresses itself is called dominant, and the allele that is masked is called recessive.

B. Law of Segregation
The two alleles of a gene separate during gamete formation. Each gamete receives only one allele. This is also called the law of purity of gametes.

C. Law of Independent Assortment
When two or more pairs of contrasting traits are studied together, the inheritance of one trait is independent of the inheritance of the other trait. This applies strictly to genes located on different chromosomes.

2. Important Genetic Terms
Gene: The basic unit of heredity that determines a characteristic.
Allele: Alternative forms of a gene (e.g., T for tall stem, t for short stem).
Homozygous: When both alleles of a gene are identical (e.g., TT or tt).
Heterozygous: When the two alleles of a gene are different (e.g., Tt).
Dominant: The allele that expresses itself in the heterozygous condition.
Recessive: The allele whose expression is masked in the heterozygous condition.
Phenotype: The observable physical or physiological characteristic of an organism.
Genotype: The genetic constitution of an organism.
Mutation: A sudden permanent change in the genetic material.
Variation: The differences that exist among individuals of the same species.

3. Monohybrid Cross (Inheritance of One Trait)
A cross between two parents differing in one contrasting trait.
Example: Tall plant (TT) crossed with Dwarf plant (tt).
F_1 Generation: All offspring were tall (Tt). The tall trait is dominant.
F_2 Generation (Selfing of F_1): On self pollination of F_1 plants (Tt × Tt), the offspring showed a ratio of 3 Tall plants to 1 Dwarf plant.
Phenotypic Ratio: 3 Tall : 1 Dwarf.
Genotypic Ratio: 1 TT (homozygous tall) : 2 Tt (heterozygous tall) : 1 tt (homozygous dwarf).

4. Dihybrid Cross (Inheritance of Two Traits)
A cross between two parents differing in two pairs of contrasting traits.
Example: Round and Yellow seeds (RRYY) crossed with Wrinkled and Green seeds (rryy).
F_1 Generation: All offspring were Round and Yellow seeds (RrYy).
F_2 Generation (Selfing of F_1): When F_1 plants (RrYy × RrYy) were selfed, the offspring showed a phenotypic ratio of 9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green.

5. Sex Determination in Humans
Humans have 23 pairs of chromosomes. Of these, 22 pairs are autosomes and 1 pair is the sex chromosome.

Females have two X chromosomes (XX). They are homogametic, producing only one type of ovum (X).

Males have one X and one Y chromosome (XY). They are heterogametic, producing two types of sperms (X and Y).

The sex of the child is determined by the type of sperm (X or Y) that fertilises the egg. If an X bearing sperm fertilises the egg, the child is female (XX). If a Y bearing sperm fertilises the egg, the child is male (XY).

6. Sex Linked Inheritance
Genes present on the sex chromosomes (specifically X chromosome) show sex linked inheritance. Males are more commonly affected because they have only one X chromosome, so a single recessive gene on it can express itself.

Haemophilia (Bleeder's Disease): A recessive X linked disorder where blood fails to clot properly due to lack of clotting factors.

Colour Blindness: A recessive X linked disorder where a person cannot distinguish between red and green colours.

IMPORTANT QUESTIONS: GENETICS

Q1. State Mendel's three laws of inheritance.

Q2. Differentiate between the following terms:
(a) Phenotype and Genotype
(b) Homozygous and Heterozygous
(c) Dominant allele and Recessive allele

Q3. With the help of a Punnett square, work out a monohybrid cross between a pure tall pea plant (TT) and a pure dwarf pea plant (tt). State the phenotypic and genotypic ratios obtained in the F_2 generation.

Q4. In a dihybrid cross between Round Yellow (RRYY) and Wrinkled Green (rryy) pea plants, what will be the phenotype of the F_1 generation? State the phenotypic ratio of the F_2 generation.

Q5. Explain the mechanism of sex determination in human beings. Who determines the sex of the child, the father or the mother?

Q6. Why are sex linked diseases like haemophilia and colour blindness more common in males than in females?

Q7. Define the following terms: Gene, Allele, Mutation, Variation.

Q8. What is the Law of Independent Assortment? Which type of cross demonstrates this law?

CHAPTER 3: ABSORPTION BY ROOTS

YouTube Lecture: https://www.youtube.com/live/sCvQZQL1fRc?si=IR_MQ3sEa60kNSFP

1. Processes Involved in Absorption

Imbibition: The absorption of water by solid particles of a substance (hydrophilic colloids), causing them to swell enormously. Example: dry seeds swelling when soaked in water.

Diffusion: The net movement of molecules or ions from a region of higher concentration to a region of lower concentration until equilibrium is reached. It does not require a membrane or energy.

Osmosis: The movement of water molecules from a region of lower solute concentration (higher water concentration) to a region of higher solute concentration (lower water concentration) through a selectively permeable membrane.

Active Transport: The movement of molecules or ions from a region of lower concentration to a region of higher concentration across a membrane, using cellular energy (ATP). This occurs against the concentration gradient.

Passive Transport: The movement of substances along the concentration gradient without the expenditure of energy. Diffusion and osmosis are examples.

2. Terms Related to Osmosis

Osmotic Pressure: The pressure that must be applied to prevent the flow of water by osmosis across a semipermeable membrane.

Root Pressure: The pressure developed in the roots of plants due to the continuous inflow of water by osmosis that helps push water up the xylem.

Turgidity: The state of a cell when it is fully swollen with water. The cell wall exerts an equal and opposite pressure (wall pressure).

Flaccidity: The state of a cell when it loses water by osmosis and becomes limp or soft.

Plasmolysis: The shrinkage of the cytoplasm away from the cell wall when a cell is placed in a hypertonic solution (solution with higher solute concentration).

Deplasmolysis: The recovery of a plasmolyzed cell when it is placed in a hypotonic solution (solution with lower solute concentration).

3. Absorption of Water by Roots

Root hairs are unicellular, thin walled extensions of the epidermal cells of roots. They increase the surface area for absorption.

Water from the soil enters the root hair by osmosis because the soil water has a lower solute concentration than the cell sap.

From the root hair, water moves through the cortical cells by osmosis, passes through the endodermis and pericycle, and finally enters the xylem vessels.

4. Ascent of Sap

The upward movement of water and dissolved minerals from the roots to the leaves through the xylem is called ascent of sap.

Forces Responsible for Ascent of Sap

- **Root Pressure:** Pushes water up (minor role).
- **Cohesion Tension Transpiration Pull Theory (Major Role):**
  - **Cohesion:** Water molecules stick to each other due to hydrogen bonding.
  - **Adhesion:** Water molecules stick to the walls of xylem vessels.

Transpiration Pull: Evaporation of water from leaf surfaces creates a suction force (tension) that pulls the entire water column upwards.

IMPORTANT QUESTIONS: ABSORPTION BY ROOTS

Q1. Define the following terms: Imbibition, Diffusion, Osmosis, Active Transport.

Q2. Explain the process of osmosis with a suitable example.

Q3. Differentiate between Turgidity and Flaccidity.

Q4. What is plasmolysis? Explain what happens when a plant cell is placed in a hypertonic solution.

Q5. Draw a labelled diagram of a root hair cell. How is it structurally adapted to absorb water?

Q6. Define ascent of sap. Name the main theory that explains the upward movement of water in plants.

Q7. Explain the Cohesion Tension Transpiration Pull theory of ascent of sap.

Q8. Differentiate between active transport and passive transport.

Q9. What is root pressure?

Q10. Name an experiment to demonstrate osmosis.

CHAPTER 4: TRANSPIRATION

YouTube Lecture: https://youtu.be/LFmIkLQwhGg?si=bicqIB1PP2-_GPH8

1. Definition of Transpiration
Transpiration is the loss of water in the form of water vapour from the aerial parts of a living plant, mainly through the stomata of leaves.

2. Types of Transpiration
Stomatal Transpiration: Water vapour lost through stomata. It accounts for about 90% of total transpiration.
Cuticular Transpiration: Water vapour lost through the cuticle layer covering the epidermis. It accounts for a small percentage.
Lenticular Transpiration: Water vapour lost through lenticels (small openings) present on the bark of woody stems. It is a very small amount.

3. Mechanism of Stomatal Transpiration (Potassium Ion Exchange Theory)
During the daytime (light), potassium ions (K^+) move into the guard cells from surrounding epidermal cells.

This increases the solute concentration in guard cells.
Water from neighbouring cells enters guard cells by osmosis. Guard cells become turgid and swell.
The thin outer wall of guard cells stretches more than the thick inner wall, causing the stomatal pore to open.
At night, potassium ions move out of the guard cells. Water follows by osmosis and the guard cells become flaccid. The stomatal pore closes.

4. Factors Affecting the Rate of Transpiration
Temperature: Higher temperature increases the rate of evaporation, so transpiration increases.
Humidity: High humidity in the air decreases the rate of transpiration because the air becomes saturated with water vapour.
Wind/Speed of Air: Wind removes the water vapour from around the leaf, increasing the rate of transpiration. Still air slows it down.
Light Intensity: Light stimulates stomata to open, increasing the rate of transpiration. In darkness, stomata close, so transpiration decreases.

5. Significance of Transpiration
It creates a transpiration pull that helps in the ascent of sap.
It has a cooling effect on the plant.
It helps in the absorption and distribution of water and dissolved minerals.

6. Important Experiments
Ganong's Potometer: Measures the rate of water uptake by a shoot. It does not measure transpiration directly but gives a comparative idea. Limitation: it measures water absorbed, not only transpiration.
Cobalt Chloride Paper: Dry cobalt chloride paper is blue. When exposed to moisture, it turns pink. It is used to compare transpiration rates from the upper and lower surfaces of a leaf. In a dorsiventral leaf, the lower surface shows more transpiration.

7. Guttation and Bleeding
Guttation: The loss of liquid water (not vapour) from the margins of leaves through special pore structures called hydathodes. Occurs in warm, humid conditions when transpiration is low.
Bleeding: The oozing out of sap (water plus dissolved substances) from a cut or injured plant part.

IMPORTANT QUESTIONS: TRANSPIRATION

Q1. Define transpiration. Name the three types of transpiration.

Q2. Explain the potassium ion exchange theory for the opening and closing of stomata.

Q3. State any three factors that affect the rate of transpiration. Explain how each factor influences it.

Q4. Give three reasons why transpiration is important for plants.

Q5. What is Ganong's potometer? What does it actually measure? State one limitation of this instrument.

Q6. How is cobalt chloride paper used to demonstrate transpiration? Which surface of a dorsiventral leaf shows higher transpiration and why?

Q7. Differentiate between transpiration and guttation.

Q8. Define bleeding in plants.

Q9. Name any two adaptations in plants to reduce transpiration.

Q10. Explain why transpiration is called a "necessary evil".

CHAPTER 5: PHOTOSYNTHESIS

YouTube Lecture: https://youtu.be/5c61BkkrnEU?si=Jvh17cLV5dMKWnzc

1. Definition and Significance
Photosynthesis is the process by which green plants synthesise food (glucose) from carbon dioxide and water using sunlight energy trapped by chlorophyll.
Significance: It is the primary source of food and oxygen for almost all living organisms. It maintains the balance of oxygen and carbon dioxide in the atmosphere.

2. Overall Equation of Photosynthesis
$6CO_2 + 12H_2O \rightarrow$ (Sunlight, Chlorophyll) $\rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$

3. Site of Photosynthesis: Chloroplast
Photosynthesis takes place inside the chloroplasts present in the mesophyll cells of leaves.
Grana: Stack of thylakoid membranes containing chlorophyll. This is the site of the light reaction.
Stroma: The fluid matrix surrounding the grana. This is the site of the dark reaction.

4. Mechanism of Photosynthesis

A. Light Reaction (Photochemical Phase)
Takes place in the grana.
Requires light energy.
Chlorophyll absorbs light energy and gets activated.
Photolysis of water occurs: water molecules split to release hydrogen ions (H^+) and oxygen (O_2). The oxygen is released as a by product.
ATP is formed (photophosphorylation).
NADPH (reduced NADP) is formed.

B. Dark Reaction (Biosynthetic Phase/Calvin Cycle)
Takes place in the stroma.
Does not require direct light but uses products of the light reaction (ATP and NADPH).
Carbon dioxide (CO_2) is assimilated and combines with hydrogen (from NADPH) to form glucose.

5. Important Experiments

Necessity of Chlorophyll: A variegated leaf is destarched and tested for starch. Only the green parts (containing chlorophyll) turn blue black with iodine, proving chlorophyll is needed.
Necessity of Light: A destarched potted plant has part of one leaf covered with black paper. After exposure to sunlight, the uncovered part turns blue black with iodine while the covered part does not.
Necessity of Carbon Dioxide: Insert a leaf of a destarched plant into a flask containing KOH (which absorbs CO_2). The leaf inside the flask shows no starch formation, while a leaf outside shows starch.
Release of Oxygen: Hydrilla/Elodea twigs are placed in water in a beaker with a funnel and inverted test tube over them. Bubbles of gas collect in the test tube. This gas relights a glowing splinter, proving it is oxygen.

6. Carbon Cycle
The carbon cycle is the circulation of carbon among the atmosphere, oceans, living organisms, and the earth's crust.
Carbon dioxide is used by plants for photosynthesis.
Carbon from plants moves to animals through food chains.
Carbon dioxide is returned to the atmosphere through respiration by plants and animals, combustion of organic fuels, and decomposition by microorganisms.

7. Adaptations in Leaves for Photosynthesis
Leaves are broad and flat to provide a large surface area for absorbing sunlight.
Leaves are thin to allow quick diffusion of gases.
Numerous stomata are present for efficient gaseous exchange.
Chloroplasts are concentrated in the upper palisade layer for maximum light absorption.

IMPORTANT QUESTIONS: PHOTOSYNTHESIS

Q1. Define photosynthesis. Write the overall balanced chemical equation for photosynthesis.

Q2. Draw a neat labelled diagram of a chloroplast and mention the site of light and dark reactions.

Q3. Differentiate between the light reaction and the dark reaction of photosynthesis (any three points).

Q4. What is photolysis of water? Why is it important?

Q5. Describe an experiment to show that light is necessary for photosynthesis.

Q6. Describe an experiment to show that chlorophyll is necessary for photosynthesis.

Q7. Explain the carbon cycle with the help of a labelled diagram.

Q8. What is destarching? Why is a plant destarched before performing a photosynthesis experiment?

Q9. List any three adaptations in leaves that make them suitable for photosynthesis.

Q10. In the Hydrilla experiment to show oxygen release during photosynthesis, why is sodium bicarbonate added to the water?

CHAPTER 6: CHEMICAL COORDINATION IN PLANTS

YouTube Lecture: https://youtu.be/XHthfFM_9Hs?si=IRfEe5MMRYksCCUO

1. Plant Growth Regulators (Phytohormones)
Plant growth regulators are small, organic molecules that control growth, development and responses in plants.

Auxins:
Promote cell elongation and growth of shoots.
Promote apical dominance (growth of the main stem tip suppresses lateral buds).
Induce root formation in stem cuttings.
Suppress premature fruit drop.
Gibberellins:
Promote stem elongation, especially in dwarf plants.
Stimulate seed germination by breaking seed dormancy.
Promote bolting (sudden elongation of stem) in rosette plants.
Cytokinins:
Promote cell division (cytokinesis).
Delay leaf senescence (ageing and yellowing).
Promote sprouting of lateral buds.
Abscisic Acid (ABA):
Acts as a growth inhibitor and a stress hormone.
Promotes seed dormancy.
Causes the closing of stomata during water stress (drought).
Promotes abscission (falling) of leaves, flowers and fruits.
Ethylene:
Promotes ripening of fruits.
Promotes yellowing and senescence of leaves.
Induces abscission of leaves.

2. Tropic Movements in Plants
Tropic movements are directional growth movements of plant parts in response to an external stimulus. They are caused by differential distribution of auxins.

Phototropism: Growth movement in response to light.
Shoots are positively phototropic (grow towards light).
Roots are negatively phototropic (grow away from light).

Geotropism (Gravitropism): Growth movement in response to gravity.
Roots are positively geotropic (grow downwards towards gravity).
Shoots are negatively geotropic (grow upwards away from gravity).

Hydrotropism: Growth movement in response to water.
Roots are positively hydrotropic (grow towards a source of water).

Thigmotropism: Growth movement in response to touch or contact with a solid object. Observed in tendrils of climbing plants that coil around a support.

Chemotropism: Growth movement in response to chemical stimulus. Example: growth of pollen tube towards the ovule during fertilisation.

IMPORTANT QUESTIONS: CHEMICAL COORDINATION IN PLANTS

Q1. Name any three plant growth regulators and state one function of each.

Q2. What is apical dominance? Which plant hormone is responsible for it?

Q3. State one function each of Gibberellins and Cytokinins.

Q4. Which plant hormone is called the stress hormone? Give two reasons.

Q5. Name the plant hormone responsible for ripening of fruits.

Q6. Define tropic movement. Explain any two types of tropic movements with suitable examples.

Q7. Differentiate between phototropism and geotropism.

Q8. What is chemotropism? Give one example.

Q9. Which hormone promotes bolting in rosette plants?

Q10. How do auxins cause phototropic curvature in shoots?

CHAPTER 7: THE CIRCULATORY SYSTEM

YouTube Lecture: https://youtu.be/bqCah5NvfTM?si=BMCocGZ4XylHcdsb

1. Composition of Blood
Blood is a fluid connective tissue composed of:
Plasma (about 55%): The liquid matrix, consisting of about 90% water with dissolved proteins, glucose, minerals, hormones, and waste products.
Red Blood Cells (Erythrocytes): Biconcave, disc shaped cells. In mammals, mature RBCs lack a nucleus to maximise space for haemoglobin. They transport oxygen from lungs to tissues.
White Blood Cells (Leucocytes): Nucleated cells that play a major role in the body's defence by fighting infections (phagocytosis and antibody production).
Platelets (Thrombocytes): Small, irregular cell fragments that help in the clotting of blood.

2. Blood Coagulation (Clotting)
When an injury occurs, platelets release thromboplastin (thrombokinase).
Thromboplastin converts prothrombin (a plasma protein) into thrombin, in the presence of calcium ions.

Thrombin then converts soluble fibrinogen (plasma protein) into insoluble fibrin threads.
Fibrin forms a mesh that traps blood cells, forming a solid clot that seals the wound.

3. Blood Groups (ABO System)
Blood groups are determined by the presence or absence of specific antigens (A and B) on the surface of RBCs and corresponding antibodies in the plasma.

Group A: Antigen A, Antibody Anti B.
Group B: Antigen B, Antibody Anti A.
Group AB: Antigens A and B, No antibodies. Known as Universal Recipient.
Group O: No antigens, Antibodies Anti A and Anti B. Known as Universal Donor.

Rh Factor: A separate antigen. If present, the blood is Rh positive (+); if absent, Rh negative (−).

4. Structure and Working of the Heart
The heart is a four chambered muscular organ.
Upper Chambers: Right Atrium and Left Atrium (receive blood).
Lower Chambers: Right Ventricle and Left Ventricle (pump blood out).
Valves: Prevent the backward flow of blood.

Systole: The phase of contraction when blood is pumped out of the chambers.
Diastole: The phase of relaxation when the chambers fill with blood.
Double Circulation: Blood passes through the heart twice in one complete circuit.
Pulmonary Circulation: Heart → Lungs → Heart (deoxygenated to oxygenated).
Systemic Circulation: Heart → Body organs → Heart (oxygenated to deoxygenated).

5. Main Blood Vessels
Vena Cava (Superior and Inferior): Bring deoxygenated blood from the body to the right atrium.
Pulmonary Veins: Bring oxygenated blood from the lungs to the left atrium.
Pulmonary Artery: Carries deoxygenated blood from the right ventricle to the lungs.
Aorta: Carries oxygenated blood from the left ventricle to all parts of the body.
Hepatic Portal Vein: Carries blood from the digestive organs to the liver before it enters general circulation.

6. Types of Blood Vessels
Arteries: Carry blood away from the heart. Thick, elastic, muscular walls. No valves.
Veins: Carry blood towards the heart. Thinner walls, wider lumen. Valves present to prevent backflow.
Capillaries: Connect arteries to veins. Walls are one cell thick to allow exchange of materials.

7. Lymphatic System
Lymph: A colourless fluid similar to blood plasma but lacking RBCs and large proteins. It transports digested fats, drains excess tissue fluid, and carries lymphocytes for defence.
Spleen: An organ that filters blood, removes old RBCs and stores blood.
Tonsils: Lymphoid tissues in the throat that trap germs entering through the mouth and nose.

IMPORTANT QUESTIONS: THE CIRCULATORY SYSTEM

Q1. Name the components of blood. State one function of each component.

Q2. Explain the process of blood coagulation (clotting).

Q3. Draw a labelled diagram of the human heart showing the chambers and major blood vessels.

Q4. Explain the concept of double circulation in humans.

Q5. Differentiate between an artery and a vein (any three points).

Q6. Name the blood vessel that:
(a) Carries oxygenated blood from the lungs to the heart.
(b) Carries deoxygenated blood from the heart to the lungs.
(c) Carries blood from the digestive organs to the liver.

Q7. Explain the ABO blood group system. Which blood group is the universal donor and which is the universal recipient?

Q8. What is the significance of the hepatic portal system?

Q9. Define systole and diastole.

Q10. State two important functions of lymph.

Q11. Why do mature mammalian RBCs lack a nucleus and mitochondria? How does this benefit them?

CHAPTER 8: THE EXCRETORY SYSTEM

YouTube Lecture: https://youtu.be/Wj8IfXfMUy0?si=mx1HL_ukJgGBibQ4

1. Introduction to Excretory Organs
Excretion is the removal of metabolic waste products from the body.
Kidneys: Primary excretory organs that filter blood and produce urine.
Lungs: Remove carbon dioxide and water vapour.
Skin: Removes sweat containing water, salts, and small amounts of urea.
Liver: Converts toxic ammonia into urea, breaks down old RBCs.

2. Parts of the Urinary System
Kidneys (2): Filter blood and produce urine.
Ureters (2): Carry urine from the kidneys to the urinary bladder.
Urinary Bladder (1): A muscular sac that temporarily stores urine.
Urethra (1): A tube that carries urine from the bladder out of the body.

3. Structure of the Kidney
A longitudinal section of the kidney shows two distinct regions.

Cortex: The outer, darker region.
Medulla: The inner, lighter region consisting of conical structures called pyramids.
Pelvis: A funnel shaped cavity that collects urine and leads to the ureter.

4. Structure and Function of the Nephron
The nephron is the functional unit of the kidney. Each kidney contains millions of nephrons.

A. Malpighian Corpuscle
Consists of the Glomerulus (a knot of capillaries) and Bowman's Capsule (a cup shaped structure surrounding the glomerulus).
Ultrafiltration: High blood pressure in the glomerulus forces water, salts, glucose, urea, and other small molecules from the blood into the Bowman's capsule. Large molecules like proteins and blood cells remain in the blood.

B. Renal Tubule
Proximal Convoluted Tubule (PCT): Selective reabsorption of useful substances (glucose, amino acids, most water, vitamins) takes place back into the surrounding capillaries.
Loop of Henle: Reabsorption of water and sodium chloride.
Distal Convoluted Tubule (DCT): Tubular secretion (removal of certain wastes, excess ions, and drugs from blood into the tubule) and further reabsorption of water.
Collecting Duct: The final concentrated urine passes into the collecting duct and then to the pelvis of the kidney.

Composition of Filtrate vs Urine: The glomerular filtrate is essentially blood plasma without proteins. Urine does not contain glucose, amino acids, or proteins but has a high concentration of urea and excess salts.

5. Blood Vessels of the Kidney
Renal Artery: Brings oxygenated blood containing waste products to the kidney.
Renal Vein: Carries filtered, deoxygenated blood away from the kidney.

IMPORTANT QUESTIONS: THE EXCRETORY SYSTEM

Q1. Name the main excretory organs in the human body and state one waste product eliminated by each.

Q2. Draw a labelled diagram of the human urinary system.

Q3. Draw a labelled diagram of a longitudinal section of the kidney.

Q4. Draw a labelled diagram of a nephron.

Q5. Explain the process of urine formation in the nephron. Name the three main steps involved.

Q6. What is ultrafiltration? Where does it occur in the nephron?

Q7. Differentiate between the composition of glomerular filtrate and urine.

Q8. Define selective reabsorption. Name two substances that are completely reabsorbed from the filtrate in a healthy person.

Q9. Name the blood vessel that brings blood to the kidney and the vessel that takes blood away.

Q10. What is tubular secretion? Name one substance secreted by this process.

CHAPTER 9: THE NERVOUS SYSTEM

YouTube Lecture: https://youtu.be/ILWpicDQqQo?si=zoNoItokB4HHuv83

1. Structure of a Neuron (Nerve Cell)
A neuron is the structural and functional unit of the nervous system.

Dendrites: Short, branched projections that receive nerve impulses from other neurons and conduct them towards the cell body.

Cell Body (Cyton): Contains the nucleus and cytoplasm. It integrates the impulses.

Axon: A single, long fibre that conducts impulses away from the cell body towards another neuron or to an effector muscle/gland.

Myelin Sheath: An insulating fatty layer around some axons that speeds up impulse conduction.

Synapse: The microscopic gap between the axon terminal of one neuron and the dendrite of the next neuron. Impulses cross the synapse through chemical substances called neurotransmitters.

2. The Brain
The brain is the main coordinating centre of the body, protected within the skull (cranium).

A. Cerebrum

The largest part of the brain.

The site of intelligence, memory, reasoning, learning, and will power.

Controls all voluntary actions.

Recieves and interprets sensory impulses (pain, touch, vision, hearing).

B. Cerebellum

Located below the cerebrum.

Primarily responsible for maintaining balance and posture of the body.

Coordinates voluntary muscle movements (e.g., walking, writing, picking up objects).

C. Medulla Oblongata

The lowest part of the brain, connecting to the spinal cord.

Controls involuntary activities like breathing, heartbeat, blood pressure, coughing, sneezing, and swallowing.

D. Thalamus

Acts as a relay centre for all sensory impulses (except smell) on their way to the cerebrum.

E. Hypothalamus

Regulates body temperature, hunger, thirst, sleep, and emotional states.

Controls the secretion of hormones from the pituitary gland.

F. Pons
Acts as a relay station between the cerebrum, cerebellum, and medulla.

3. The Spinal Cord
The spinal cord extends from the medulla oblongata and runs down through the vertebral column.
White Matter: Outer region containing bundles of myelinated nerve fibres (axons).
Gray Matter: Inner butterfly shaped region containing cell bodies of neurons.
Functions: Conducts impulses to and from the brain. Serves as the centre for reflex actions.

4. Reflex Action and Reflex Arc
Reflex Action: A rapid, automatic, involuntary response to a stimulus that occurs without conscious thought. It is protective in nature.
Reflex Arc: The simple neural pathway along which impulses travel during a reflex action.
Pathway: Receptor (sensory organ) → Sensory Neuron → Spinal Cord (Relay Neuron) → Motor Neuron → Effector (muscle/gland).
Natural Reflex: Inborn, not learned (e.g., blinking, sneezing, withdrawing hand on touching a hot object).
Acquired/Conditioned Reflex: Learned through experience or practice (e.g., salivating at the sight or smell of food, cycling).

5. The Eye
The eye is the organ of sight.
Main Parts and Functions
Sclera: The tough, white, outer protective layer.
Cornea: The transparent front part of the sclera that allows light to enter and refracts (bends) it.
Choroid: The middle layer, rich in blood vessels, that nourishes the retina.
Retina: The innermost layer containing light sensitive receptors. Rods are responsible for vision in dim light (black and white). Cones are responsible for colour vision.
Lens: A transparent, crystalline structure behind the pupil that fine focuses light onto the retina.
Pupil: The opening in the centre of the iris through which light enters.
Iris: The coloured part surrounding the pupil that controls the size of the pupil.
Accommodation: The process by which the shape of the lens changes to focus on near or distant objects on the retina.

Stereoscopic Vision
Having two eyes (binocular vision) provides overlapping fields of view, allowing the brain to perceive depth and judge distance.

Eye Defects and Corrective Measures
Myopia (Short Sightedness): Can see near objects clearly but not far objects. The image forms in front of the retina due to an elongated eyeball. Corrected using a concave (diverging) lens.
Hypermetropia (Long Sightedness): Can see far objects clearly but not near objects. The image forms behind the retina due to a shortened eyeball. Corrected using a convex (converging) lens.
Presbyopia: Age related condition where the lens loses elasticity, making it hard to focus on near objects. Corrected using a bifocal lens.
Astigmatism: Blurred vision caused by an uneven curvature of the cornea. Corrected using a cylindrical lens.

Cataract: The lens becomes cloudy or opaque, leading to loss of vision. Treated by surgical removal of the lens and implantation of an artificial lens.

6. The Ear
The ear is the organ of hearing and balance.
Outer Ear: Includes the pinna (collects sound waves) and the auditory canal (directs sound to the eardrum).
Middle Ear: The eardrum (tympanum) vibrates in response to sound. Three tiny ear ossicles (malleus, incus, stapes) amplify these vibrations and transmit them to the oval window.
Inner Ear: The cochlea is a coiled structure containing a fluid and hair cells that convert vibrations into nerve impulses for hearing. The semicircular canals are loop shaped structures responsible for the sense of body balance (equilibrium).

IMPORTANT QUESTIONS: THE NERVOUS SYSTEM

Q1. Draw a labelled diagram of a neuron. State the function of dendrites and axon.

Q2. What is a synapse? How does an impulse travel across a synapse?

Q3. Draw a labelled diagram of the brain (external view) and state one function each of Cerebrum, Cerebellum and Medulla Oblongata.

Q4. Differentiate between voluntary and involuntary actions. Give one example of each.

Q5. Explain the reflex arc with the help of a labelled diagram. Why are reflexes important?

Q6. Distinguish between natural reflex and acquired reflex with examples.

Q7. Draw a labelled diagram of the human eye.

Q8. Explain the following defects of the eye and state how they are corrected:
(a) Myopia
(b) Hypermetropia

Q9. What is accommodation of the eye? How is it achieved?

Q10. Draw a labelled diagram of the human ear. State the role of the cochlea and semicircular canals.

Q11. Explain stereoscopic vision. Why is it advantageous?

Q12. State the functions of the thalamus and hypothalamus.

CHAPTER 10: THE ENDOCRINE SYSTEM

YouTube Lecture: https://youtu.be/dQJg4lSbnPo?si=CybSpVeNMfZx4Iri

1. Endocrine vs Exocrine Glands
Endocrine Glands: Ductless glands that secrete their products (hormones) directly into the bloodstream. Hormones act on target organs at a distance.
Exocrine Glands: Glands that have ducts. They secrete their products (enzymes, sweat, saliva) through these ducts onto a surface or into a cavity.

2. Key Endocrine Glands, Hormones and Functions

A. Pituitary Gland (Master Gland)
Located at the base of the brain.
Growth Hormone (GH): Stimulates overall body growth.
Tropic Hormones (TSH, ACTH, FSH, LH): Control the functioning of other endocrine glands (thyroid, adrenal, gonads).
ADH (Antidiuretic Hormone/Vasopressin): Reduces water loss by increasing water reabsorption in the kidneys.
Oxytocin: Stimulates contractions of the uterus during childbirth.

B. Thyroid Gland
Located in the neck region.
Thyroxine: Regulates the basal metabolic rate (BMR) of the body, controls growth and development.
Deficiency (Hyposecretion) causes Goitre (swelling of the thyroid), Cretinism in children (stunted growth and mental retardation).
Excess (Hypersecretion) causes Exophthalmic Goitre (bulging eyes, increased metabolism).

C. Adrenal Gland
Located above the kidneys.
Adrenaline (from Medulla): The 'emergency' or 'fight or flight' hormone. It increases heart rate, blood pressure, breathing rate, and releases glucose for instant energy.
Cortical Hormones (from Cortex): Regulate salt and water balance in the body and help in carbohydrate metabolism.

D. Pancreas (Islets of Langerhans)
Located in the abdomen behind the stomach. Acts as both an exocrine and endocrine gland.
Insulin (Beta cells): Lowers blood glucose level by promoting its conversion into glycogen. Deficiency causes Diabetes Mellitus.
Glucagon (Alpha cells): Raises blood glucose level by converting stored glycogen back into glucose.

3. Feedback Mechanism (Negative Feedback)
A regulatory mechanism that maintains hormonal balance. For example, the pituitary secretes TSH, which stimulates the thyroid to produce thyroxine. When the level of thyroxine in the blood increases, it inhibits the release of TSH from the pituitary, which in turn reduces thyroxine production.

IMPORTANT QUESTIONS: THE ENDOCRINE SYSTEM

Q1. Differentiate between an endocrine gland and an exocrine gland.

Q2. Why is the pituitary gland called the master gland? Name any two hormones secreted by it and state their functions.

Q3. Name the hormone secreted by the thyroid gland. State the effects of its hyposecretion and hypersecretion.

Q4. Which hormone is known as the emergency hormone? Name the gland that secretes it and list its effects on the body.

Q5. Name the hormones secreted by the Islets of Langerhans. Explain the role of insulin and glucagon in regulating blood sugar.

Q6. Explain the negative feedback mechanism with reference to TSH and thyroxine.

Q7. What is the function of ADH? Which gland secretes it?

Q8. Name the hormone that promotes contraction of the uterus during childbirth.

Q9. State any three differences between nervous control and hormonal control.

Q10. A person shows symptoms of bulging eyes and increased heart rate. Name the condition and the hormone responsible.

CHAPTER 11: THE REPRODUCTIVE SYSTEM

YouTube Lecture: https://youtu.be/lfU-C0pDRlo?si=-qEYnNQr6QGTONkb

1. Male Reproductive System
  Testes (paired): Produce the male gametes (sperms) and the male hormone Testosterone.
  Epididymis: A coiled tube on top of each testis where sperms mature and are temporarily stored.
  Vas Deferens (Sperm Duct): A tube that carries sperms from the epididymis to the urethra.
  Accessory Glands:
    Seminal Vesicles
    Prostate Gland
    Cowper's Glands
    Their secretions form the liquid portion of semen, which provides nutrition and a fluid medium for sperms.
  Urethra: Common passage for urine and semen.

2. Female Reproductive System
  Ovaries (paired): Produce the female gametes (ova/eggs). Also secrete the hormones Oestrogen and Progesterone.
  Fallopian Tubes (Oviducts): Funnel shaped tubes that catch the ovum released from the ovary. They carry the ovum towards the uterus. Fertilisation takes place here.

Uterus (Womb): A muscular, pear shaped organ where the fertilised egg implants and the embryo develops.
Vagina: The birth canal. It receives the sperms during mating.

3. Fertilisation, Implantation and Development
Fertilisation: The fusion of the nucleus of a sperm with the nucleus of an ovum to form a zygote. Occurs in the fallopian tube.
Implantation: The process by which the developing zygote/embryo attaches itself to the endometrium (inner lining) of the uterus.
Gestation: The period of development of the embryo in the uterus.
Parturition: The act of giving birth to a fully developed foetus.

4. The Placenta
The placenta is a disc shaped organ formed by both embryonic and maternal tissues.
Functions of the Placenta:
Nutrition: Transfers nutrients (glucose, amino acids) from the mother's blood to the embryo.
Respiration: Supplies O_2 from the mother to the foetus and removes CO_2 from the foetus to the mother.
Excretion: Removes nitrogenous wastes (urea) from the foetus into the maternal blood.
Endocrine Function: Produces hormones like progesterone, oestrogen and hCG (human chorionic gonadotropin) to maintain pregnancy.
Role of Amniotic Fluid: The embryo floats in amniotic fluid inside the amniotic cavity. It protects the embryo from physical shock, prevents it from drying out, and allows some movement.

5. Menstrual Cycle
The cyclic changes in the ovaries and the uterus of a female during her reproductive years, controlled by hormones. The cycle is roughly 28 days.
Menstrual Phase (Days 1 to 5): The thick, spongy lining of the uterus (endometrium) breaks down and is shed along with blood through the vagina. This is menstruation.
Follicular Phase (Days 6 to 13): A follicle in the ovary matures under the influence of FSH. The maturing follicle secretes Oestrogen, which causes the endometrium to thicken.
Ovulation (Day 14): The mature follicle ruptures, releasing the ovum into the fallopian tube.
Luteal Phase (Days 15 to 28): The empty follicle develops into the Corpus Luteum, which secretes Progesterone. Progesterone maintains the endometrium for implantation. If fertilisation does not occur, the corpus luteum degenerates, progesterone level drops, and the cycle restarts with menstruation.

6. Sex Hormones and Their Roles
Testosterone: Responsible for primary and secondary sexual characters in males (deepening of voice, growth of facial hair, sperm production).
Oestrogen: Responsible for primary and secondary sexual characters in females (breast development, widening of hips) and repair of the endometrium after menstruation.
Progesterone: Prepares the endometrium for implantation of the fertilised ovum and maintains pregnancy.

7. Twin Types
Identical (Monozygotic) Twins: Formed from a single fertilised egg that splits into two at an early stage. They are genetically identical and always of the same sex.
Fraternal (Dizygotic) Twins: Formed when two separate eggs are fertilised by two separate sperms simultaneously. They are genetically different and may be of the same or different sex.

IMPORTANT QUESTIONS: THE REPRODUCTIVE SYSTEM

Q1. Draw a labelled diagram of the male reproductive system.

Q2. Draw a labelled diagram of the female reproductive system.

Q3. Name the site of fertilisation and the site of implantation in humans.

Q4. State any four functions of the placenta.

Q5. Explain the role of amniotic fluid during embryonic development.

Q6. Describe the phases of the menstrual cycle. What happens if the ovum is not fertilised?

Q7. Name the hormones secreted by the ovaries and state one function of each.

Q8. List any three secondary sexual characters in males controlled by testosterone.

Q9. Differentiate between identical twins and fraternal twins.

Q10. Define the following terms: Fertilisation, Implantation, Gestation, Parturition.

CHAPTER 12: POPULATION

YouTube Lecture: https://youtu.be/2QKio4tL1GY?si=IwGyKZNtKLXc09q3

1. Key Demographic Terms
Demography: The statistical study of human populations, including their size, structure, and distribution.
Population Density: The number of individuals living per unit area (e.g., per square kilometre).
Birth Rate (Natality): The number of live births per thousand individuals per year.
Death Rate (Mortality): The number of deaths per thousand individuals per year.
Growth Rate: The net increase in population size, calculated as (Birth Rate − Death Rate) plus net migration.

2. Reasons for Population Explosion in India
Decline in death rate due to improved medical facilities, better sanitation, and control of epidemics.
High birth rate due to early marriage, lack of education, desire for male child, and religious/social customs.
Increase in average life expectancy.

3. Problems Caused by Population Explosion
Unemployment: Jobs become scarce compared to the number of job seekers.

Over exploitation of Natural Resources: Increased demand leads to deforestation, water scarcity, and depletion of fossil fuels.
Low Per Capita Income: Economic growth is outpaced by population growth, reducing the share of national income per person.
Price Rise (Inflation): Demand for goods and services exceeds supply, causing prices to increase.
Pollution: Increased waste generation leads to air, water, and soil pollution.
Unequal Distribution of Wealth: The gap between the rich and the poor widens.

4. Methods of Population Control
Surgical Methods (Sterilisation):
Vasectomy: In males, a small segment of the vas deferens is cut and tied. This prevents sperms from reaching the urethra.
Tubectomy: In females, a small segment of the fallopian tubes is cut and tied. This prevents the ovum from meeting the sperm.
Barrier Methods: Use of condoms, diaphragms, etc. to prevent sperm from reaching the ovum.
Chemical/Hormonal Methods: Use of oral contraceptive pills, implants, etc.
Intrauterine Devices (IUDs): Devices placed inside the uterus to prevent implantation.

IMPORTANT QUESTIONS: POPULATION

Q1. Define the following terms: Demography, Population Density, Birth Rate, Death Rate.

Q2. Give any three reasons for the rapid rise in human population in India.

Q3. Explain any four problems caused by population explosion.

Q4. What is vasectomy? How does it help in population control?

Q5. What is tubectomy? How does it help in population control?

Q6. Differentiate between birth rate and death rate.

Q7. List any two non surgical methods of contraception.

Q8. How does population growth lead to environmental pollution?

CHAPTER 13: POLLUTION

YouTube Lecture: https://youtu.be/lfU-C0pDRlo?si=kB7rSHVFhlzL3vjY

1. Types of Pollution and Major Pollutants

Air Pollution
Sources: Vehicular emissions, industrial smoke, burning of garbage, brick kilns.

Major Pollutants: Carbon monoxide (CO), Sulphur dioxide (SO_2), Nitrogen oxides (NOx), suspended particulate matter (dust, smoke).

Water Pollution
Sources: Household detergents, sewage, industrial discharge (heavy metals, chemicals), oil spills.
Major Pollutants: Organic waste (increases BOD), chemicals (fertilisers, pesticides), pathogens.

Soil Pollution
Sources: Excessive use of chemical fertilisers and pesticides, urban commercial and domestic waste, untreated industrial waste.
Major Pollutants: DDT and other pesticides, non biodegradable plastics, heavy metals.

Noise Pollution
Sources: Motor vehicles, industrial establishments, construction sites, loudspeakers.
Major Pollutants: Unwanted, excessive, and disturbing sounds measured in decibels (dB).

Radiation Pollution
Sources: X rays, radioactive fallout from nuclear power plants and nuclear weapons testing.

Biomedical Waste
Sources: Hospitals, nursing homes, clinics.
Major Wastes: Used and discarded needles, syringes, soiled dressings, expired medicines. These require special treatment to prevent spread of infections.

2. Biodegradable and Non Biodegradable Wastes
Biodegradable Wastes: Wastes that can be decomposed by the natural action of microorganisms (bacteria, fungi) over time. Examples: Paper, vegetable and fruit peels, agricultural waste, animal dung.
Non Biodegradable Wastes: Wastes that cannot be broken down by microorganisms and persist in the environment for many years. Examples: Plastics, glass, metals, Styrofoam, synthetic fibres. Pesticides like DDT are chemically stable and bioaccumulate.

3. Effects of Pollution

Greenhouse Effect and Global Warming
The increase in concentration of greenhouse gases (CO_2, methane, water vapour) traps heat radiated from the Earth's surface.
This leads to a rise in global average temperature (global warming), causing polar ice melt, rise in sea level, and climatic changes.

Acid Rain
Gases like SO_2 and NOx mix with water vapour in the atmosphere to form sulphuric and nitric acids.
These fall as acid rain, damaging forests, acidifying lakes and rivers, and eroding buildings and monuments (like the Taj Mahal).

Ozone Layer Depletion
The ozone layer in the stratosphere protects life on Earth by absorbing harmful ultraviolet (UV) radiation from the sun.

Chlorofluorocarbons (CFCs) used in refrigerants and aerosols react with and destroy ozone molecules, depleting the ozone layer. This increases the amount of UV reaching the Earth, causing skin cancer, cataracts, and damage to ecosystems.

4. Measures to Control Pollution
Use of unleaded petrol and CNG (Compressed Natural Gas) in automobiles.
Switching off vehicle engines at traffic signals to save fuel and reduce emissions.
Social forestry and afforestation (planting more trees).
Setting up sewage treatment plants to treat waste water before discharge.
Banning of polythene bags and single use plastics.
Promoting organic farming to reduce chemical fertiliser and pesticide use.
Following Euro and Bharat Stage vehicular emission standards.
"Swachh Bharat Abhiyan" (Clean India Mission): A national campaign to create a clean India through proper sanitation and waste management.

IMPORTANT QUESTIONS: POLLUTION

Q1. Define pollution. Name any four types of pollution and give one source for each.

Q2. Differentiate between biodegradable and non biodegradable wastes. Give two examples of each.

Q3. What is biomedical waste? Why should it be handled and disposed of carefully?

Q4. Explain the greenhouse effect and how it contributes to global warming. Name two greenhouse gases.

Q5. What is acid rain? Name the two gases primarily responsible for it. State two harmful effects of acid rain.

Q6. Explain the importance of the ozone layer. How do CFCs cause ozone layer depletion? State two effects of ozone depletion on human health.

Q7. Suggest any five measures that can be taken to control pollution.

Q8. What is the Swachh Bharat Abhiyan? What is its main objective?

Q9. Why is CNG considered a cleaner fuel for vehicles?

Q10. What is social forestry? How does it help in controlling pollution?